6CL6 RF Driver Stage

Updated from 12BY7A version -- single-ended, drives push-pull PA via Tr2

Why 6CL6

The 6CL6 is essentially a 6.3 V heater version of the 12BY7A in every parameter that matters here. Both are 9-pin miniature beam pentodes designed for video / wideband / RF service. Key parameters are within a few percent of each other.

The schematic, operating point, and all component values are **unchanged from the 12BY7A version**. The only difference is the heater wiring (now 6.3 V across pins 4-5 instead of 12.6 V).

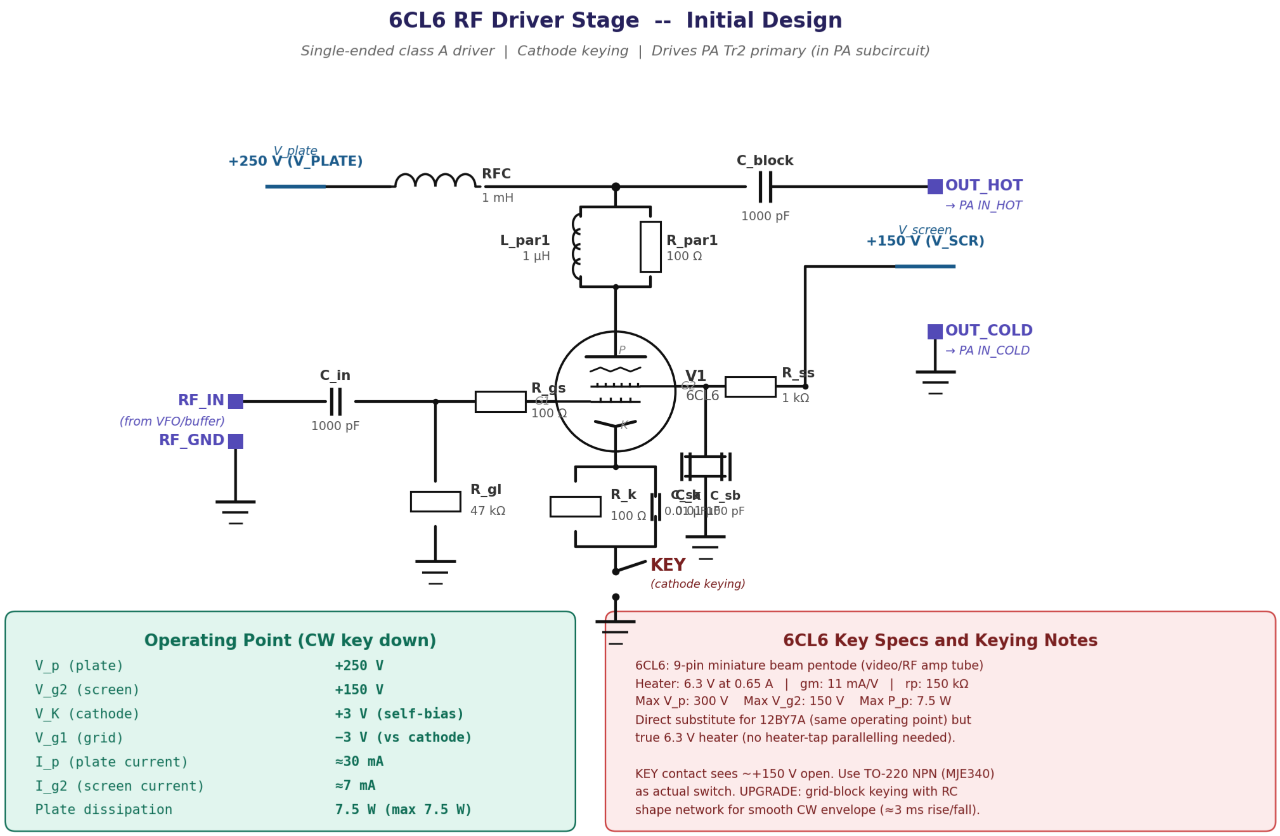

Figure 1 -- 6CL6 driver stage. Identical topology and component values to the previous 12BY7A version, just the tube swapped. Drives the PA Tr2 primary through C_block.

Topology Reminder

Single-ended class A driver feeding the PA via the Tr2 phase-splitting transformer (located inside the PA subcircuit). The output of this stage is a single-ended RF signal at the plate, which Tr2 converts to balanced push-pull drive for the two 6146B grids.

Plate is parallel-fed: RFC to B+ keeps the DC supply isolated from the AC signal path, and C_block prevents the +250 V plate voltage from reaching the PA input. Plate parasitic suppressor (L_par1 ■ R_par1) damps any VHF resonance through the tube's plate-grid capacitance.

Screen is bypassed at the G2 pin with parallel HF + VHF caps directly to chassis (the lesson we learned the hard way on the PA). Without these the screen impedance contributes a feedback path that causes oscillation.

Component Values

Ref	Value	Purpose
C_in	1000 pF	Grid coupling from VFO/buffer
R_gl	47 kΩ	Grid leak (return to ground)
R_gs	100 Ω / 0.5 W	Grid stopper at G1 pin (VHF damping)
R_k	100 Ω / 1 W	Cathode self-bias resistor (V_K ≈ 3 V)
C_k	0.01 µF / 50 V	Cathode bypass (Xc=1.1Ω at 14 MHz)
R_ss	1 kΩ / 2 W	Screen stopper
C_sa	0.01 µF / 200 V	Screen HF bypass at G2 pin
C_sb	100 pF / 200 V	Screen VHF bypass at G2 pin
RFC	1 mH	Plate RF choke (B+ feed)
L_par1	1 µH	Plate parasitic suppressor (■ R_par1)
R_par1	100 Ω / 2 W	Plate parasitic damper
C_block	1000 pF / 500 V	Plate DC blocking cap to PA input
KEY	NPN switch	Cathode keying transistor (e.g., MJE340)

Heater Wiring Note

6CL6 heater is between pins 4 and 5. With 6.3 V supply, wire directly (unlike the 12BY7A which would need pins 4 and 5 paralleled if running on 6.3 V).

All 6146B and 6CL6 heaters can share the same 6.3 V supply line, but use generous wire gauge -- a pair of 6146B (3.75 A total) plus a 6CL6 (0.65 A) totals 4.4 A on the heater bus. Twisted pair to keep magnetically-induced hum out of the RF signal path.

Next Steps

1. Verify tube.lib has a 6CL6 model. If not, the 12BY7A model in the library should be a close enough substitute for simulation purposes (they're essentially the same tube). Alternative: use a 6AG7 model and adjust operating point slightly.

2. Build as Driver_Subcircuit.sch with ports RF_IN, RF_GND, OUT_HOT, OUT_COLD. V_plate and V_screen as ideal sources for simulation.

3. Test standalone with 1 V / 14.2 MHz sine into RF_IN, looking at plate voltage swing. Should see roughly 50-100 V pk-pk at the plate into a reasonable load.

4. Integrate with PA subcircuit. The driver's OUT_HOT and OUT_COLD connect directly to the PA's IN_HOT and IN_COLD (which are the Tr2 primary terminals, with R13 providing DC ground reference).

5. Once driver+PA chain works in simulation, add the keying shaper circuit (separate transistor + RC network between morse key and the KEY line shown here).